Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models the motion of the particle by forming a second order differential equation. (must have correct terms but allow sign errors)	AO3.3	M1	$M\frac{d^2x}{dt^2} = -4M\frac{dx}{dt} - 8Mx$ $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0$
	Obtains correct differential equation	AO1.1b	A1	$\lambda^2 + 4\lambda + 8 = 0$ $\lambda = -2 \pm 2i$
	Forms and solves auxiliary equation for 'their' D.E.	AO1.1a	M1	Complex roots ⇒ General solution is of the form:
	States a correct form of the general solution for 'their' auxiliary solution. (ft only if both M1 marks have been awarded)	AO1.1b	A1F	$x = Ae^{-2t}\cos(2t + B)$ $\dot{x} = -2Ae^{-2t}\cos(2t + B)$ $-2Ae^{-2t}\sin(2t + B)$
	Uses initial conditions to find arbitrary constants for 'their' solution	AO1.1a	M1	$\dot{x}(0) = 0$ so, $-2A\cos(B) - 2A\sin(B) = 0$
	Obtains correct value for one of 'their' constants (ft only if all M1 marks have been awarded)	AO1.1b	A1F	$\Rightarrow \tan B = -1$ $\Rightarrow B = -\frac{\pi}{4}$
	Obtains correct value for both of 'their' constants (ft only if all M1 marks have been awarded)	AO1.1b	A1F	$x(0) = 1$ $\Rightarrow \frac{A}{\sqrt{2}} = 1$
	Uses 'their' model to describe the motion of the particle either as a written description or shown on a clearly labelled graph.	AO3.4	A1F	$\Rightarrow A = \sqrt{2}$ $x = \sqrt{2}e^{-2t}\cos(2t - \frac{\pi}{4})$ ∴ the particle oscillates about O, with period π seconds and decreasing amplitude.

(b)	Refines their DE model to account for altered resistive force by introducing a new	AO3.5c	B1	$\frac{d^2x}{dt^2} + \alpha\frac{dx}{dt} + 8x = 0$

coefficient for $\frac{dx}{dt}$			Critical damping ⇒ $\lambda^2 + \alpha\lambda + 8 = 0$ must have equal roots $\alpha^2 = 32$ $\alpha = 4\sqrt{2}$ Resistive force should have magnitude $4\sqrt{2}Mv$
Uses or states condition for critical damping	AO1.2	B1	
Deduces value for coefficient of $\frac{dx}{dt}$	AO2.2a	R1	
States resistive force	AO3.4	B1	
			Total 12 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)(i)	Forms a differential equation for the foxes.	AO3.3	B1	$\frac{dy}{dt} \propto x \Rightarrow \frac{dy}{dt} = kx$
	Forms a differential equation for the rabbits.	AO3.3	B1	$\frac{dy}{dt} = 20,\ x = 200 \Rightarrow k = 0.1$ $\frac{dy}{dt} = 0.1x$ $\frac{dx}{dt} = 1.2x - 1.1y$
	Differentiates 'their' equation that contains y and obtains expression with at least two terms correct.	AO1.1a	M1	$y = \frac{1.2x}{1.1} - \frac{1}{1.1}\frac{dx}{dt}$
	Formulates a second order linear differential equation.	AO3.1a	M1	$\frac{dy}{dt} = \frac{1.2}{1.1}\frac{dx}{dt} - \frac{1}{1.1}\frac{d^2x}{dt^2}$
	Obtains roots of auxiliary equation for 'their' second order differential equation.	AO1.1a	M1	$0.1x = \frac{1.2}{1.1}\frac{dx}{dt} - \frac{1}{1.1}\frac{d^2x}{dt^2}$ $\frac{d^2x}{dt^2} - 1.2\frac{dx}{dt} + 0.11x = 0$
	States correct general solution. FT provided all M marks have been awarded	AO1.1b	A1F	$\lambda^2 - 1.2\lambda + 0.11 = 0$ $\lambda = 0.1$ or 1.1 $x = Ae^{0.1t} + Be^{1.1t}$
	Uses initial population to find equation linking constants for their general solution.	AO3.4	M1	$t = 0,\ x = 80$ $A + B = 80$
	Obtains initial rate of change for rabbits from 'their' DE and differentiates and obtains a second equations for A and B from 'their' general solution.	AO3.4	M1	$t = 0, \frac{dx}{dt} = 74$

	Finds A and B and states the model for the number of rabbits.	AO1.1a	M1	$74 = 0.1A + 1.1B$ $74 = 0.1(80 - B) + 1.1B$ $B = 66$, $A = 14$ $x = 14e^{0.1t} + 66e^{1.1t}$
(ii)	Substitutes 0.7 and obtains approximately 160. CAO	AO3.4	A1	$14e^{0.07} + 66e^{0.77} = 157.6$
(b)	States a suitable refinement about the fact that an increased rabbit population will require more food supply or other valid refinement	AO3.5c	B1	Take account of the food available for the rabbits as this may limit population growth.
				Total 11 marks